Revisiting inter-AS IP Spoofing Let the Protection Drive Source Address Validation

Yihao Jia*†‡, Ying Liu†‡, Gang Ren†‡✉, Lin He*†‡
‡Institute for Network Sciences and Cyberspace, Tsinghua University, Beijing 100084, China
*Department of Computer Science and Technology, Tsinghua University, Beijing 100084, China
†Tsinghua National Laboratory for Information Science and Technology, Beijing 100084, China

***Abstract*—IP spoofing, which is prevalently used for anonymity and reflection attacks, has shown increasing destructive power in recent years. Although certain source address validation solutions have been standardized by the Internet Engineering Task Force, few networks are willing to adopt them in view of the deficiency of deployment benefits. Actually, all the source address validation solutions face the problem of a lack of deployability. In this paper, we summarize the key points describing deployability and propose a new security service—inter-autonomous-system (AS) Source Address Protection (*iSAP*). Technically, by increasing the possibility of keeping the source address belonging to one AS from being the victim of reflection flooding, *iSAP* improves the deployers ability to prevent IP spoofing and increases incremental deployability. In reality, such a service can also be regarded as a new profit opportunity for ASes and it could progress gradually once it is well commercialized. Based on simulations with real Internet topology data, the results illustrate that *iSAP* can protect ASes from being reflected with only a few deployers, exhibiting a high potential to mitigate reflection flooding with modest resource consumption.**

I. Introduction

As is well known, distributed denial-of-service (DDoS) [1] attacks are among the most prevalent forms of attack hazarding Internet security. According to the report from Arbor Networks [2], the traffic peak of DDoS attacks have surged from 100 Gbps in 2010 to at least 800 Gbps in 2016. IP spoofing, which is one of the most critical dependencies of a DDoS attack, can be classified into two types: d-DDoS for destination-orientated and s-DDoS for source-orientated [3] attacks. The d-DDoS may attack a target with arbitrary source addresses to remain anonymous, whereas the s-DDoS, which is also defined as a reflection attack, sends packets with the target's source addresses to innocent destination hosts, whose replies may then flood the target. Although both the source and destination addresses are candidate victims of IP spoofing, in typical cases, it is the "source" that suffers the most. Practically, because many protocols above the IP layer will reply with packets of several times the requests size, the flooding power can be easily amplified by this vulnerability based on reflection. In fact, the DDoS attack records have been continually broken by these s-DDoS attacks—300 Gbps in 2013 by DNS reflection and 400 Gbps in 2014 by NTP reflection, and the statistics in [2] show that s-DDoS attacks are much more diverse when compared to d-DDoS, and comprise the majority of the DDoS attacks.

Source address validation architecture (SAVA) [4] was proposed to defend against IP spoofing. SAVA divides the mission into three levels with different granularity: the access network, intra-autonomous system (AS), and inter-AS. The first two levels focus on the validation of the packets that originate in the AS (internal orientation), whereas the last level considers the packets originating from other ASes (external orientation). Extension mechanisms such as [5] and Ingress/Egress Filtering [6], representing the first two levels, have been standardized by the Internet Engineering Task Force (IETF) and its Source Address Validation Improvements (SAVI) working group. However, these methods are beneficial only for the rest of the Internet rather than for the deployers themselves, and also present risks for the deployers when servicing their own users if the configurations have errors. For this reason, many ASes gradually lose interest in deploying these methods [7].

The distributed nature of DDoS suggests that a distributed mechanism is necessary for a successful defense, and thus large scale deployment is especially critical to combat such a distributed threat. As inter-AS defense is the only level that prevents IP spoofing from outside an AS, the mechanisms in this level can undoubtedly upgrade the deployer ASes' network security by mitigating extraneous spoofing and bring them more benefits than those in the other two levels. For this reason, proposing an efficient inter-AS defense method is absolutely vital to realize wide deployment, and IP spoofing will be significantly crippled if an increasing number of ASes decide to get involved.

In this study, we propose an inter-AS Source Address Protection service—*iSAP*. Owing to the ubiquity and detrimental effects of s-DDoS attacks, *iSAP* takes the prevention of reflection flooding as the breakthrough of the incentives design, and selects deployability as the main objective of the methodology study. The contributions of this paper can be summarized as follows. First, we present three deployability-oriented policies as the guidance for the method design. These policies pinpoint the key factors that should be considered when constructing a deployable method. Second, we define the concept of source address protection to describe the defense of reflection flooding and propose the *iSAP* service. *iSAP* meets all the deployability requirements well, and maintains scalabil-

✉ Gang Ren (rengang@cernet.edu.cn) is the corresponding author.

978-1-5090-6468-7/17/$31.00 © 2017 IEEE

ity by the decoupled structure at the same time. Third, to better assess the deployability, we introduce new metrics to quantize the benefits for deployers and evaluate its effectiveness. By extensive simulations with real Internet data, *iSAP* shows a high potential to prevent s-DDoS attacks with modest resource consumption, exhibiting decent efficiency in providing security service of source address protection.

The rest of this paper is organized as follows. Section II covers design orientation. Then, we describe *iSAP* in detail in Section III. This is followed by the evaluation in Section IV. Section V discusses related work, and concluding remarks are presented in Section VI.

II. Design Orientation

Although several methods to defend against IP spoofing were proposed in the last decade, only a limited number have been put into use within the AS, and no mechanism has been widely deployed across AS in actual operating networks. By probing the roots of this phenomenon, we found that it is the limitation of deployability that hinders their development. For this reason, we analyze three factors related to the dilemma behind these methods' deployment in this section, and regard them as the main aspects that guide our design.

A. Incentives to Deployer

For most source address validation researches, especially the internal-based ones, method designers usually overemphasize the "validation" and global efficiency, and ignore the "benefit" for these practical deployers. However, as for deployer ASes, operators of these ASes are often reluctant to complicate their networks' configurations if no extra benefits are provided. In most cases, the deficiency of incentives to deployers significantly impedes the deployment. In this regard, we introduce two policies that are focused on providing further incentives.

1) Source address protection: To inspire the deployers, "benefit" must be clearly defined. In view of the ubiquity and detrimental effects of s-DDoS intrusion, a source-oriented protection mode could be a more important and practical motivation toward security improvement. For this reason, we use the "protection" to approach the "benefit" when ASes deploy related mechanisms. In general, the "protection" in this paper is defined as **the probability of keeping source addresses of one AS from being the victims of inter-AS reflection flooding**, i.e. the probability that the address belonging to one AS can be inhibited from being spoofed and employed by others. Specially, we use three parameters to depict an inter-AS reflection attack—source AS s, destination AS d, and victim AS v. The source AS is the one which the attacker hosts reside in and the spoofing traffic stems from, while the destination AS is the AS which the spoofing traffic reach to and the reflectors reside in. Victim AS is the one whose IP address is spoofed as the source address by the above traffic and then flooded by these innocent reflectors. Then, if the traffic spoofing v from s to d can be filtered during AS path (s...d) forwarding, such filtering can be regarded as a successful "protection" for AS v. Regrettably, however, these existing methods still lack the capacity to offer ASes such protection during source address validation.

2) Source address protection as a service: Once source address security could be designed as a protection, such protection could be further treated as a service and charged for by ASes who offer them. Specifically, ASes owning the capability of discerning and filtering spoofing can offer the victim AS source address protection, and the victim AS may be willing to pay if it can be certainly protected from s-DDoS flooding. In reality, there would be a great many victim ASes suffering from reflection attacks; therefore, capable ASes may be more willing to provide new services for new profits. Optimistically, the mechanisms involved in this service may also be iteratively upgraded to offer more professional security service during the service's commercialization.

B. Incremental Deployability

From a holistic perspective, three points should be considered to ensure that ASes are continuously willing to keep the method deployed within the inter-AS environment.

1) Effectiveness for partial deployment: To persuade the early deployers, especially in an inter-AS area, the mechanisms must work in a partial deployment environment. However, for methods such as Source Address Validity Enforcement (*SAVE*) [8] and *BGP Selection Notice* [9], the validation can only be executed when all routers in the topology install the new mechanisms. For this reason, methods that require global deployment at the beginning are not practical in the inter-AS environment.

2) Effectiveness for nonadjacent environment: In addition to the partial deployment, methods are supposed to work when deployers are nonadjacent. For example, BGPsec can prevent a route from hijacking only when **all** ASes on the path are secure, while the security of the inter-AS routing will not be improved unless the vast majority of ASes deploy that method [10]. Therefore, source address validation methods are always expected to be efficient, even in a sparse and nonadjacent deployment environment.

3) Effectiveness for sustaining deployment: Another important factor for ASes continuously adopting the method is to ensure that the source address security will monotonously increase as more ASes implement such a method. If the ASes are not able to get any further benefits from their adoption at a certain deployment ratio, such growth will gradually stop, and the mechanisms are also difficult to be further supported by the community in the future.

C. Deployment Overhead

To clear the obstacles further, the overhead of running the mechanism should be controlled. Two policies could be introduced in our design philosophy.

1) From online to offline: We use the term "online" to represent the validation requiring additional cryptography-based computation for each packet in real time, and "offline" to represent the validation requiring no additional computation for each packet in real time, based on rules generated in

advance. In terms of the overhead, an offline validation method is always preferable than the online ones. Taking another online validation method—BGPsec as an example, since the cost of updating the hardware might unduly frustrate the early investor, the upgrade from BGP to BGPsec is considered to be sluggish and could take an extremely long period [11]. For this reason, it is suggested that methods which require online validation be avoided in order to keep the overhead at a controllable level.

2) From routers to server: Methods that involve a protocol update usually require function operations by network nodes. Instead, for services that offer specific features, the operations can be transferred to a third party. In other words, similar to the relying party to the system of RPKI (resource public key infrastructure), the functions and overhead can be handed over to a third server when security is regarded as a service. Therefore, with regard to the source address validation, if rules can be generated and updated in a third server, the routers will be free from taking the control plane responsibilities.

III. Inter-AS Source Address Protection

Based on the analysis in Section II, we instantiate a concrete method to provide the inter-AS source address "protection service" and name it *iSAP* (inter-AS Source Address Protection). As implied, *iSAP* emphasizes the benefits for deployers by protecting their source address from being spoofed, i.e. **by increasing the probability of keeping their source address from being the victim of reflected flooding**. For this reason, we regard the *"protection"* of source address as a specific description of *"validation"*, and use them alternately to describe the methods that prevent IP spoofing.

A. Architecture

Owing to the power-law theorem of the Internet [12], routing-based validation is supposed to be an offline validation method with high filtering efficiency [13]. For deployability, *iSAP* follows the routing-based mechanisms to build the core part of the filtering framework.

Two basic characters are involved in the *iSAP* service—*iSAP* subscriber AS and *iSAP* provider AS. The *iSAP* provider AS is the one that offers the source address protection service, whereas the *iSAP* subscriber AS is the AS that requests the service. In this architecture, one subscriber AS may request source address protection from multiple provider ASes, while one provider AS may provide multiple subscriber ASes with such a service. This is a many-to-many architecture. In addition, to provide such protection, the *iSAP* subscriber AS should deploy an *iSAP* subscriber server, whereas an *iSAP* provider server is needed in the *iSAP* provider AS. Figure 1 describes a simple example of an *iSAP* architecture. For conciseness, we omit the "*iSAP*" in front of the subscriber and provider server, and draw them in pink and blue, respectively, in the figures throughout the rest of the paper.

Since multiple AS border routers (ASBRs) can be abstracted as different interfaces of a large-scale one, for brevity, we suppose each AS deploys one ASBR to abstract the multiple

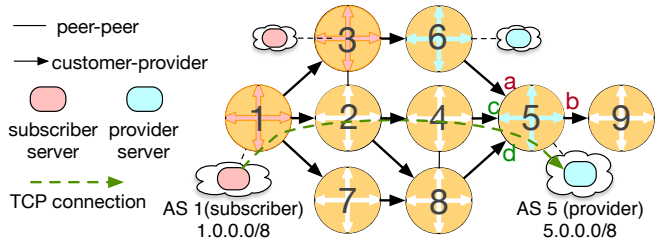


Fig. 1. *iSAP* System Architecture

ASBRs as its network gateway, and the ASes are interconnected by these abstracted ASBRs. In Figure 1, AS_x owns the IP address space of $x.0.0.0/8$. AS_1 and AS_3 are the subscriber ASes, whereas AS_5 and AS_6 are provider ASes. In this section, the process of service subscription from AS_1 to AS_5 is described in detail. Suppose that, somehow, the AS_5 has realized that traffic from AS_1 can only be transmitted by neighbor AS_4 or AS_8; then, the packets with source address belonging to $1.0.0.0/8$ and coming from interface a (AS_6) and b (AS_9) must be spoofing and should be filtered out.

In general, since only the subscriber AS knows the route through the provider AS under the BGP inter-domain routing environment, AS_5 cannot filter the $1.0.0.0/8$-based spoofing traffic for AS_1 unless AS_1 informs the feasible upstream ASes to AS_5 itself. In *iSAP*, the subscriber and provider servers undertake the responsibility of delivering such information. Architecturally, the process by which subscriber ASes request source address protection from providers can be divided into the following steps.

1. The subscriber server will capture all the routing selection results from its ASBR; (routing selection results are presented at an AS granularity as AS path).
2. For each target provider AS, the subscriber server will pick up all their candidate upstream ASes before reaching them from the routing selection results, forming preliminary rules for the protection of the addresses they own.
3. The subscriber server will then connect to all provider servers that provide it with source address protection. Data transfer in these connections includes the preliminary rules and its own address space, forming the integral rules in all its provider servers.
4. Based on the network topology of the provider ASes, the provider servers will further transform the integral rules into configurable incoming tables and install them in their ASBRs for spoofing filtering.

In order to represent this system clearly, we first focus on the ASBR of both subscriber and provider ASes. The inner details of the ASBR will elucidate the different responsibilities of each role.

1) ASBR for subscriber: For the ASBR of a subscriber AS, there is no special update as against a normal one, and the mission for a subscriber ASBR is still packet forwarding. However, to inform the subscriber server of its own routing selection results, the only extra mechanism for this subscriber

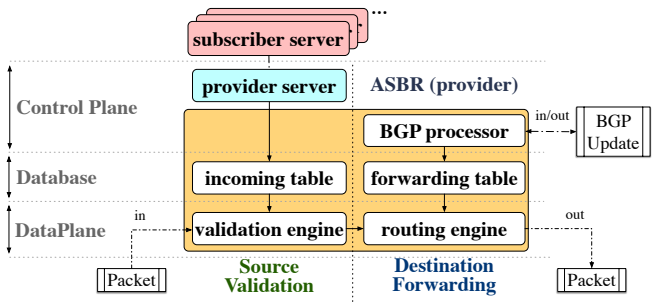


Fig. 2. ASBR for provider

AS is to establish an e-BGP session between the ASBR and subscriber server. The details of this e-BGP session will be provided in the next subsection.

2) ASBR for provider: To protect the address of subscriber ASes, the ASBRs of the provider ASes must be updated to filter the spoofed packets. Therefore, additional missions and modules for packet source address validation will be installed in provider ASBRs. To avoid the disorder of inter-domain routing as much as possible, we suggest decoupling the source validation from the destination forwarding at the same routing table. Since the ongoing BGPsec has already caused difficulty during its deployment [11], integrating a new immature function into BGP will certainly risk the stability and reachability of the huge distributed system. For this reason, we parallelize the source (address)-based validation and the destination (address)-based forwarding of providers' ASBRs to increase the flexibility and isolate the interference from each side (Figure 2). Compared to the process of forming the forwarding table, the incoming table used for source address validation will be fed by the provider server, whereas the materials in the provider server will be supplied by several subscriber servers.

B. Rules Initialization

In this section, we detail the process of establishing the validation rules in the initial stage of *iSAP*. Figure 3 depicts the structure of the subscriber and provider servers and the entire working flow of *iSAP*, while the non-shaded part of Figure 4 describes the intermediate states of this initial stage. For clarity, we draw the "daemon" and "table" in these servers with green and yellow, respectively.

Consider AS_1 and AS_5 in Figure 1 as an example for detailed illustration. First, the ASBR of AS_1 establishes an eBGP session to the eBGP daemon. This session is configured in such a way as to ensure that all the selected routes will be exported to the subscriber server. Then, the selected routes will be recorded in a RIBin-like table, as shown by $TableA$ in Figure 4. After that, the rules extractor—$daemon_B$ will execute the following algorithm to generate the preliminary rules. It should be noted that all the ASes in the inter-domain area are assumed to run BGP accurately. Cases for BGP hijacking will be explained at the end of this section.

$providerList$: the set of all provider ASes
$AS.upstreamList$: the set of legitimate upstream ASes for a given ***provider AS***
for all $AS \in providerList$ **do**
  $AS.upstreamList \leftarrow \emptyset$
  for all $row \in RIBinLikeTable$ **do**
    if $AS \in row.ASpath$ **then**
      $index \leftarrow row.ASpath.getIndex(AS)$
      $upstreamAS \leftarrow row.ASpath[index - 1]$
      $AS.upstreamList.append(upstreamAS)$
    end if
  end for
end for

The process of finding all feasible upstream ASes for a given provider (AS) involves enumerating all possible choices from the selected routing table. In Figure 1, we suppose that the only routes arriving or passing through AS_5 are route *ASpath1,2,4,5* and *ASpath1,2,8,5,9*. By traversing the entire RIBin-like table, we can find that only AS_4 and AS_8 are the feasible upstream nodes before reaching AS_5. The non-shaded part of $TableB$ in Figure 4 shows the results of this algorithm.

After the preliminary rules are established in the subscriber server of AS_1, they would be sent immediately to the provider server of AS_5 by TCP connection. In addition, AS_1 must offer AS_5 the address space it owns, while AS_5 should verify the address space by a mechanism such as RPKI. In such communications, TLS or other encryption techniques should be added to prevent attacks.

Then, in $TableC$, integral rules (non-shadow part of $TableC$) will be formed in the AS_5 provider server. According to the network topology interconnecting AS_5's neighbors, AS_4 and AS_8 refers to the interface c and d in the AS_5 border router. Therefore, the integral rules will be transformed into $TableE$ as the general rules for source address protection. Here, we name $TableE$ as the incoming table for the comparison of the forwarding table in the providers' ASBR.

When there is an s-DDoS attack on AS_1, AS_1 will dynamically invoke the protection service from AS_5 through the communication between their subscriber and provider servers. After that, the provider server in AS_5 will install $TableE$ in all its ASBRs as the filtering incoming table, and after the attack, the protection service would be repealed from AS_1. Apparently, the dynamic invocation feature not only empowers *iSAP* with the flexibility for charge and operation, but lowers the possibility of misconfiguration and potential false positives.

C. Policies for Dynamic Routing

Although dynamic invocation of *iSAP* mitigates its risks to a certain degree, the routes passing through a provider AS may be changed during an invocation. To overcome this drawback, two problems must be solved to ensure that the rules for source address validation are consistent with the dynamic routing. The first problem is the dynamic adjustment of the rules during dynamic routing, while the second problem is to ensure that the dynamic control plane message will not be mistakenly filtered after the routes are dynamically changed.

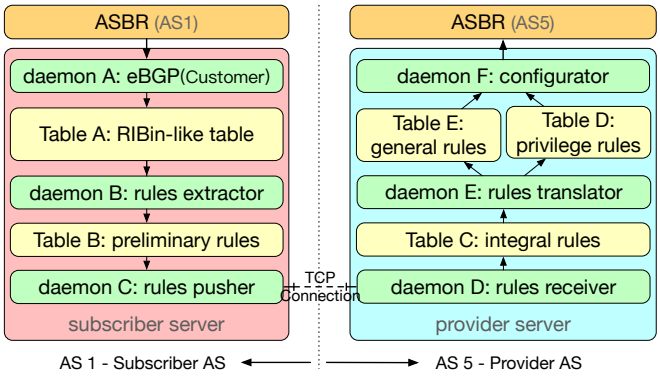


Fig. 3. Server structure and working flow

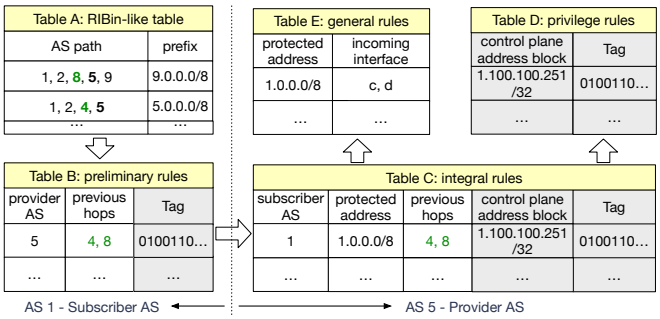


Fig. 4. Tables for intermediate states

1) Dynamic adjustment: In the example shown in Figure 1, suppose that somehow the ASBR of AS_2 breaks down, and AS_1 decides to adopt route *ASpath1,3,6,5* rather than any other path to forward the traffic that is going to AS_5. In this situation, AS_5 must immediately adjust the rules for $1.0.0.0/8$ to a new legitimate interface a and temporarily close c and d. However, since each route change related to AS_1 will trigger a new BGP update at AS_1, such adjustment could be accommodated by an extra logic according to the new BGP update. The pseudo code, which describes the algorithm for such dynamic adjustment in $daemon_B$ is given below.

```
for all AS ∈ providerList do
    if AS ∈ BGPupdate.ASpath then
        index ← BGPupdate.ASpath.getIndex(AS)
        upstreamAS ← BGPupdate.ASpath[index − 1]
        AS.upstreamList.append(upstreamAS)
    end if
end for
```

For each BGP update announcing a new route, it will be checked whether it's "AS path" contains a provider AS. If it contains one, then the upstream AS in this route will be added as a legitimate one in the validation rules. Besides, it should be noted that algorithm given above only shoulders the responsibility of adding new legitimate upstream ASes during protection invocation in an emergency circumstance, and the deletion of the obsolete upstreams will still be handled in a similar way as that of the initial stage. Moreover, a new BGP update in protection invocation period may also refer to a route withdraw, and such withdraw can be regarded as upstream deletion as well, and handled similar to the initialization.

2) Privilege for control plane message: Considering that the connection between the subscriber and provider servers is based on a TCP session—a protocol above the IP layer, a control plane message may be mistakenly filtered in a protection invocation period since the ASBRs are not able to distinguish a control plane message depending on the original IP header only. In the previous case, it must be ensured that the control plane message from AS_1 can pass the validation at the interface a of the AS_5 border router after the route change. For this reason, to ensure the arrival of the control plane message, *iSAP* is not able to operate in a mere routing-based way. An extra privilege label—the labeling-based mechanism—should be introduced as an auxiliary within the IP header.

To avoid the superfluous overhead, *iSAP* selects "tag", rather than the cryptographic MAC (Message Authentication Code), to compose the label. Before service invocation, each subscriber and provider AS will negotiate a pseudo-random number generation algorithm to generate a tag periodically for privilege use. Then, during attacks and service invocation, the control plane message triggered by dynamic routing will be tagged as the negotiated sequence at the host of the subscriber server. After that, when these control plane packets arrive at a provider AS, the ASBRs of this provider AS can differentiate these packets from spoofing packets by the tags validity.

The space for accommodating the tag is quite different in IPv4 and IPv6. Since the size of the update message for dynamic routing can be restricted in a non-fragment packet, the "Identification" (16-bit) and "Fragment Offset" (13-bit) seems suitable for sequence rewriting (29-bit in total). For IPv6, such tags can be perfectly accommodated in an adequate bit zone of a new IPv6 extension header.

The shaded parts of the tables in Figure 4 describe the privilege tags in *iSAP* system. During the negotiation, the subscriber server will inform the providers the designate source address space that should be validated by the tag sequence. Then, in the provider server, the shaded part of $TableC$ can be split into privilege rules as given in $TableD$. Each time when protection service is invoked, both $TableD$ and $TableE$ will be installed on the providers' ASBRs, and during validation $TableD$ will be considered prior to $TableE$.

Suppose that the IP address of the subscriber server is 1.100.100.253, and the tag sequence negotiated between AS_1 and AS_5 is "0100110...". Therefore, when the route from AS_1 to AS_5 shifts to *ASpath1,3,6,5*, the AS_1 subscriber server will capture the related BGP update and trigger the control plane update message with the tag "0100110..." at the IP header of the packet. When the packet arrives at interface a of the AS_5 border router, the rules described by $TableE$ are not appropriate, because the update messages always lag behind the routing changes. Therefore, the ASBR should first search $TableD$ and validate the packet by tag "0100110..". Finally, when the communication is complete, $TableE$ in AS_5 will be updated, and the rules for the incoming table will be restored

to accommodate these new routes.

3) Security risks assessment: As mentioned above, since update messages always lag behind the routing changes, the short interval between changes and adjustments may still cause packet loss due to the obsolete rules. However, such damages caused by this hysteretic nature can only occur during the invocation period, i.e. flooding period. In such periods, the packet loss seems to be a normal case since the network nodes of victims are unable to process the traffic at that volume, while protocols like TCP may always help to restore the communications by mechanisms of congestion control. Compared to the detriment of s-DDoS flooding, the collateral damages of UDP connections are worth taking, since damage at such short intervals may far surpass that loss.

Besides, since tag sequence is relatively static when compared to the MAC, there would be a scruple whether the privilege for *iSAP* tag can be utilized by those unauthorized attackers. The countermeasure of mitigating such risk is to narrow its valid scope. According to the rules described in $TableD$, the privilege for tag is strictly restricted to the control plane addresses, i.e. a small address block. In other words, for a target outside the block, the privilege tag cannot be abused. Besides, even if the attackers attempt to flood a control plane server by brute-force, the threats for control plane servers may also be easily frustrated by a high-frequency tag refreshment or the popularization of IPv6 and its roomy extension header.

D. Platform Design

Although the mechanisms described above give *iSAP* the ability to provide source address protection, it is still necessary to build an indexing platform for providers to achieve collaborations with the subscribers. Here, *iSAP* introduces a third-party platform such as *PeeringDB* [14] to complete this task. If an AS has deployed *iSAP* as a provider, it can register the service related information on such third-party platforms. Then, for ASes that deploy *iSAP* as subscribers and are seeking source address protection, they can retrieve the providers' information in this third-party and reach a consensus with some of them. For these provider-subscriber peers that have reached a cooperation consensus, configurations such as certifications for secure communication would be shared before the initialization of the mechanism.

E. Deployment Policy

The most critical problem for *iSAP* is how to choose the early deployers. Intuitively, since small parts of ASes occupy most of the inter-domain traffic [12], ASes that are located at the core of the Internet are more valuable as providers in *iSAP* system. For this reason, the selection of early deployers should never be random; it should include well-connected ASes like the Tier 1s and Tier 2s. However, as proven in [15], it is difficult to even approximate the optimal set of early deployers, and practically, ASes are unlikely to deploy the method in the theoretically optimal order. Therefore, a suboptimal policy is enough to guide and evaluate the *iSAP* deployment. The metric we use to approach the ideal policy is the order of AS transit degree [16]. In short, the AS transit degree, which was proposed in [17] as a key metric for AS relationship detection, depicts the number of ASes that were observed as receiving transit paths through a given AS. For this reason, ASes that own a higher transit degree should be more recommended and appropriate to be the early deployers in *iSAP* deployment. The effectiveness of such a policy will be proved in the "Evaluation" Section IV.

F. Discussion

Before moving to the evaluation, we briefly discuss whether *iSAP* matches the orientations in Section II, and whether it would be perturbed by routing hijacking.

1) Embodiment of design orientation: For properties, the incentive requirements drive the *iSAP* to provide "protection" for deployers themselves, and put it into a service mode as a new profit point. In terms of s-DDoS, no free-rider is involved. Since *iSAP* only protects the subscribers who adopt the methods, legacy ASes are unable to get free benefits as they used to from previous methods. Thus, the obstacle that restrains the mechanisms from incremental deployment has been cleared under this methodology. In addition, *iSAP* works well in its partial deployment, without asking for adjacent environments. When more ASes are ready to provide such security service, there is more possibility for the spoofing traffic to be transmitted and filtered by the provider routers, and more victims may be willing to get themselves involved. Besides, since all rules are formed at the third servers, routers may focus on the offline validation with relatively lightweight overheads.

2) Status about route hijacking: As mentioned above, since the source (address) based validation and destination (address) based forwarding are two parallel compositions, the risks for routing will not affect the operations of packet validation. Nevertheless, the process forming the validation rules is still based on BGP—the de facto inter-domain routing protocol. In other words, the rules for validation in *iSAP* always obey the choices of the routing protocol. For this reason, *iSAP* would not crash during route hijacking, but dynamically "adapts" it. After the adaption, no traffic will be dropped by *iSAP* when the hijack remains undetected. After the hijack, the *iSAP* will immediately revert to the original rules as in normal routing.

IV. Evaluation and Analysis

As discussed above, *iSAP* meets the requirements for deployment from a holistic and macroscopic perspective. In this section, we evaluate the "deployability" of *iSAP* by quantizing the "benefits" for these practical deployers. First, we illustrate the scheme of simulating the attack and defense. Then, metrics embodying the "benefits" are introduced with statistical results and analysis. Finally, we discuss the overhead throughout the entire validation process. For conciseness, we denote the universal set of all ASes existing with the number of n by N, and the set of all provider deployers by M.

1) Hypothesis: The first assumption is that any arbitrary AS shares the same probability of being the source, destination, or victim AS. Mathematically, let p_i^S, p_i^D, and p_i^V depict the independent probabilities of an AS i being the source, destination, or victim AS. Then $\forall i$, $0 \leq p_i^S, p_i^D, p_i^V \leq 1$, $p_i^S = p_i^D = p_i^V$, $\sum_i p_i^S = \sum_i p_i^D = \sum_i p_i^V = 1$. Another hypothesis is about the volume of the spoofing traffic. Since the spoofing traffic for any target reflector is highly stochastic and is decided by the attackers subjective consciousness, for the sake of simplicity, we suppose that the volumes for spoofing traffic between any source and destination AS shares the same weight. That is, $\forall a, i \in$ source AS and $\forall b, j \in$ destination AS, and the spoofing volume $Vol(a, b) = Vol(i, j)$.

2) Datasets and tools: The inter-domain graph we used in this evaluation is a real Internet topology obtained from CAIDA (Center for Applied Internet Data Analysis). To restore such AS-level graphs, we adopted the dataset of "AS Relationships" on "2016-05-01" [18], and adopted the "prefix-ASN" from the "RouteViews Prefix to AS Mappings" on the same date [19]. Besides, to capture the route for any two AS pairs, we introduced C-BGP [20] as a sophisticated simulator to find the AS path in the graph.

3) Deployer selection policy: To measure the effectiveness for each metric comprehensively, *iSAP* should be evaluated under different provider deployment ratios. Initially, we let $M = \emptyset$, and at each sampling point we strategically selected a legacy AS in M and calculated the metrics. For the first policy, we randomly selected the legacy AS to be the *iSAP* provider. We named it "random" strategy. Then, in the second round, we always selected the legacy AS with the largest transit degree, instead of a random one, as our "ideal" strategy. Besides, considering that it is insignificant to persuade stub-ASes to provide *iSAP* service, sampling points would be more practical in a partial deployment environment.

4) Comparison: Based on the design orientations, methods that not support incremental deployment are not comparable in our evaluations, and methods without explicit benefits for deployers should also be omitted. For this reason, we select "*DISCS*" [3] and "*SPM*" [21] as the comparison in metrics evaluation considering their influence in this research area and the compatible assessment model. Besides, since "*DISCS*" shares the same filtering principles with"*SPM*" but with optimized policies and implementation, we treat them as the same model in the comparison evaluations.

A. Evaluations

As described in Section II, in previous researches, the evaluations usually focused on, or even overemphasized the efficiency for the whole community, at the price of sacrificing the "benefits" for deployers. Besides, since *iSAP* protects a given subscriber from being the victim of s-DDoS attack, the metrics for the whole Internet do not make much sense. More importantly, the efficiency at an extensive deployment environment seems to be meaningless if the method cannot be accepted by deployers themselves. Therefore, to better assess the deployability, the metrics in *iSAP* evaluation will be devised and optimized for practical deployers.

For deployers, the greatest impetus that stimulate ASes to adopt a method is the "benefits" they get. Since there are two different characters in *iSAP* service, "benefits" for these characters would not be the same. For providers, "benefits" usually comes from the payment of subscribers. However, ASes are willing to subscribe to the service only when they can get considerable "benefits" in such subscription. Therefore, to inspire the providers, it must be proven that subscribers are offered adequate protection against flooding attack, and after that, the subscription may stimulate more top ASes to deploy as providers and get extra profits in this new kind of business. For subscribers, the "benefits" mainly stem from the "protection" they get from *iSAP* service, i.e. **the ability of inhibiting their addresses from being spoofed and employed by others, thus keeping themselves away from reflection flooding**. For this reason, the metrics described below will be more focused on quantizing the "benefits" for these subscribers. Without loss of generality, the deployers describing the deployment ratio refer to the provider ASes, while the metrics below are focused on the evaluation of subscriber ASes.

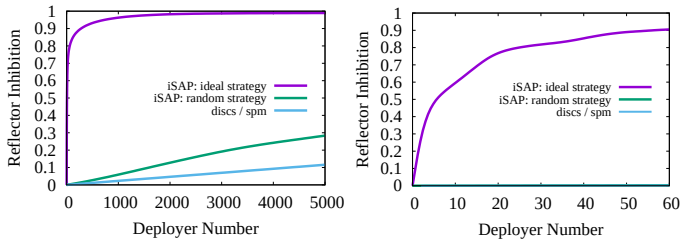
1) Average Reflector Inhibition ($\bar{I}$): Suppose the botnets owned by an attacker pervade all source ASes. Then, for a given victim, spoofing traffic can stem from all source ASes to a target reflector residing in the destination AS. Average reflector inhibition ($\bar{I}$) is introduced to assess the defense efficiency of such flooding. Macroscopically, for an arbitrary victim v and set M, $\bar{I}$ shows the impossibility, for an arbitrary AS, of being the reflector AS and responding to the spoofing traffic. Mathematically, the spoofed packets that originate from source AS s to destination AS d are denoted by $spf(s, d, v)$, with the source address belonging to the victim AS v. Let $filter\{M, spf(s, d, v)\}$ depict the result of whether AS v could be immune from a specific $spf(s, d, v)$ under the protection of M. Then $filter$ can be given as follows:

$$filter\{M, spf(s,d,v)\} = \begin{cases} 1, & \text{if } M \text{ can filter } spf(s,d,v) \\ 0, & \text{otherwise} \end{cases}$$

Since the number that the routes arrive at a certain destination AS d is $n-1$, for a given victim AS v and the destination AS d, $I(v|d, M) = \frac{\sum_s filter\{M, spf(s,d,v)\}}{n-1}$. Therefore, $\bar{I}(v|M)$ can be formulated as the average of $I(v|d, M)$ as:

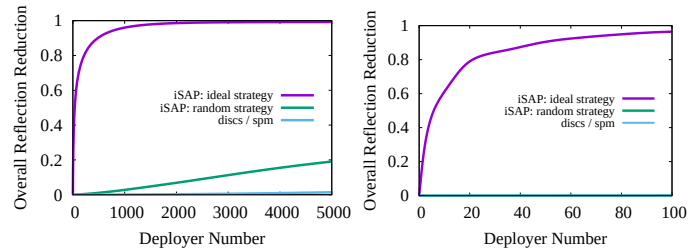
$$\bar{I}(v|M) = \frac{\sum_d I(v|d, M)}{n}$$

By traversing v, we obtain the results shown in Figure 5. According to Figure 5a, $\bar{I}$ shows a monotonic increasing tendency in inhibiting reflection attacks. On comparing the values in "*DISCS / SPM*", both the strategies of *iSAP* offer subscribers higher "Average Reflector Inhibition" for an arbitrary sampling point. Besides, as depicted in Figure 5b, for the ideal strategy, $\bar{I}$ bursts rapidly at the early deployment stage with more than 90% under 60 deployers. In other words, for any given AS d, the spoofing traffic attacking v and utilizing d as the reflector



(a) extensive deployment (b) early stage deployment

Fig. 5. Average Reflector Inhibition ($\bar{I}$)



(a) extensive deployment (b) early stage deployment

Fig. 6. Overall Reflection Reduction (R)

will be weakened by these deployers by about 90%. In this regard, *iSAP* benefits from the Internet power-law theorem, and offers subscribers significant benefits in preventing reflection flooding, once the providers are deliberately selected.

2) Overall Reflection Reduction (R): The metric, overall reflection reduction (R), is described as follows. For an arbitrary victim v and deployer set M, $R(v|M)$ indicates the reduction in reflection of the spoofing traffic from any source AS s to any destination AS d by the defense from M. Since there are $n*(n-1)$ routes from s to d, $R(v|M)$ can be described as follows, based on the definitions in $\bar{I}$.

$$R(v|M) = \sum_{s,d} \frac{filter\{M, spf(s,d,v)\}}{n*(n-1)}$$

Figure 6 illustrates the results obtained by traversing v. As shown in Figure 6a, R presents a similar tendency as that in $\bar{I}$. The results shown in Figure 6b reveal that with the ideal strategy, R surges to 60% with only 12 deployers, and for a certain victim, 95% of the spoofing traffic will be prevented before reflection by only 90 deployers. That is to say, if the ideal strategy is applied, *iSAP* can provide subscribers considerable protection against s-DDoS attacks with only a few deployers from both average ($\bar{I}$) and overall (R) perspectives.

3) Bandwidth Economization (E): Except for the benefits for deployers themselves, *iSAP* may economize the bandwidth for the whole community when compared to traditional end-to-end validation methods. Theoretically, *iSAP* filters spoofing traffic on intermediate ASes as they pass by, and therefore, subsequent ASes may avoid serving invalid traffic and keep them away from consuming resources. Therefore, since end-to-end methods can only filter spoofing at the destination AS, packets which can be validated by methods like "*DISCS / SPM*" may be dropped earlier in *iSAP* environment. To simulate the process of bandwidth economization, we extract the inter-AS hop count of the packets before they are filtered as an abstraction of the bandwidth utility.

Compared to *DISCS / SPM*, figure 7 shows the bandwidth economization for the on-path filtering feature of *iSAP*. For the "ideal" strategy, *iSAP* saves more than 50% bandwidth with only 80 deployers, and the value rises to 60% with less than 800 deployers. Besides, even for the "random" strategy, 20% of the spoofing traffic will be filtered earlier with merely

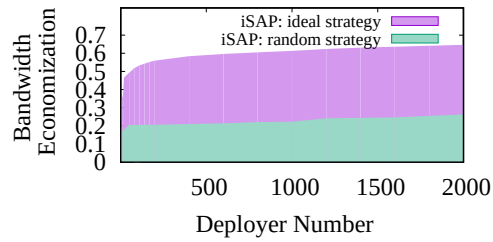


Fig. 7. Bandwidth Economization (E)

a few deployers. In this regard, *iSAP* improves the filtering efficiency by avoiding serving spoofing.

B. Overhead Analysis

1) Control plane overhead: The overheads for the initiation of rules and the adjustment for dynamic routing are different. Since the initial rules for source address protection are established before service invocation, it can be regarded as off-line and $o(0)$ overhead. For dynamic routing, only one update message will be triggered when new BGP update arrives for each routing change, and therefore, the overhead during the service invocation is up to a constant $o(1)$.

2) Data plane overhead: The data plane overhead in *iSAP* lies in the validation at the providers' ASBRs. Suppose that the overhead for packet forwarding in an ASBR is $o(L)$ (L refers to the number of prefixes listed in the routing table), and all the ASes have already deployed *iSAP* as subscribers. Since no cryptography is introduced in *iSAP* system, the overhead during filtering may stem only from table searching. Considering that only two more tables are involved in providers' ASBRs, the overhead of data plane is limited to $o(2L) \approx o(L)$.

V. Related Work

The vulnerabilities of address security motivated many proposals for inter-AS source address validation. In general, the methods can be primarily classified into two types: cryptography-based validation and routing-based validation. Due to restrictions in length, we only discuss the main representatives for each kind of proposal, and analyze the contrast when compared to *iSAP*.

A. Labeling-based Validation

Labeling-based validations usually introduce an extra cryptography system, and the packets in this system will be verified

by the cryptography label related to their source addresses. *Passport* [22], *DISCS* [3], and *SPM* [21] are representatives of the methods of this type. However, such methods usually require online computation for the entire traffic an AS receives, and hardware updates supporting these massive computations are quite costly when compared to the benefits they provide. Moreover, in most cases, packets will only be validated at the destination AS; therefore, it is difficult to exploit the power-law theory for improving the efficiency using the cryptography method. In *iSAP*, although the "tag" label is also introduced as the auxiliary, such a label can only work 1) in certain small address blocks, 2) during service invocation only for privilege use. Meanwhile, since the non-cryptographic "tag" ensure that *iSAP* always validates packets offline, rather than online, software upgrades are enough for *iSAP* validation.

B. Routing-based Validation

Unicast Reverse Path Forwarding (*uRPF*) [23], which utilizes the forwarding table as the filter rules for source validation, shows high potency when deployed at the edge of the Internet. However, substantially, *uRPF* is an internal defense method, rather than external. Considering that stub-AS occupies more than 85% of the entire inter-domain unites [24], it is unrealistic to persuade all these AS to deploy such methods without any further benefit. Besides, although methods like *IDPF* [25] can be deployed at the core of the Internet, the asymmetrical routing makes it impracticable to deploy in a real environment. In addition, due to the requirement of global deployment, methods like *SAVE* [8] and *BGP Selection Notice* [9] are unrealistic to be accepted in the inter-domain area. In summary, the deficiency of deployability restricts previous routing-based methods from being applied in the distributed inter-domain environment.

VI. Conclusion

Although the inter-AS IP spoofing is a severe threat to Internet security, none of the solutions is practically deployable. By analyzing the reason of this phenomenon, we believe that the deployability must be considered while designing a new method from scratch. On the one hand, we consider the incentives to the deployer as the most important index, since it determines to what degrees the ASes are willing to deploy the methods. On the other hand, to ensure that ASes have a sustaining will for adopting this mechanism through the whole stages of deployment, the method must be devised to support incremental deployment from the beginning. For this reason, we propose to emphasize the source address "protection", rather than "validation" as a new cognition for research in this area. Since "protection" represents the benefits for deployers, ASes have more motive for adopting such method.

In this paper, we proposed a routing-based, labeling-assisted inter-AS source address validation method—*iSAP*. As proven above, *iSAP* protects the address space that the subscriber owns from being spoofed by others, thus reducing the probability of being the victim of reflection flooding. From the design perspective, the role partition for different deployers (provider and subscriber) makes *iSAP* practically more deployable and flexible when compared to other solutions, while the introduction of the service mode also provides decent benefits for both the subscribers (in the form of security) and providers (in the form of revenue). Although models are simplified when describing attacks and defenses in evaluations, results show that *iSAP* have sufficient potential to mitigate large-scale s-DDoS flooding with only a few providers, exhibiting decent filtering efficiency with modest resource consumption.

Most importantly, we explored the possibility of providing Source Address Protection as a Service (SAPaaS). Although specific charging models are skipped in this paper due to the complicated economic and business model, (e.g., providers may charge their subscribers according to the amount of the spoofing packets they filtered), *iSAP* is capable enough to serve as the first attempt at such a service. Besides, as mentioned above, the *iSAP* solution is a totally decoupled structure, which means that the mechanisms in *iSAP* could be easily superseded by new methods with higher efficiency, or be gradually enriched as technologies improve. Nevertheless, how to persuade the top ASes to invest and provide the service at the very beginning still seems to be a tough problem. We suggest that governments and industry groups can focus on regulatory efforts and financial incentives to convince large ISPs to become the early providers in their countries, and the community will be inspired if ASes at the core of the Internet decide to provide such services.

Acknowledgment

This work is supported by the National Natural Science Foundation of China (NSFC) under grant No. 61772307 & No. 61402257, National Basic Research Program of China (973 Program) under grant No. 2009CB320500 & No. 2009CB320501, and Tsinghua University Self-determined Project under grant No. 2014z21051.

References

[1] S. Yu, "Distributed denial of service attack and defense," in *Springer Briefs in Computer Science*, 2014.

[2] D. Anstee, P. Bowen, C. Chui, and G. Sockrider, "12th worldwide infrastructure security report," *Arbor Network Inc.*, 2017.

[3] B. Liu and J. Bi, "Discs: A distributed collaboration system for inter-as spoofing defense," in *Parallel Processing (ICPP), 2015 44th International Conference on*. IEEE, 2015, pp. 160–169.

[4] J. Wu, J. Bi, X. Li, G. Ren, M. Williams, and K. Xu, "A Source Address Validation Architecture (SAVA) Testbed and Deployment Experience," RFC 5210, Jun. 2008.

[5] J. Wu, J. Bi, M. Bagnulo, F. Baker, and C. Vogt, "Source Address Validation Improvement (SAVI) Framework," RFC 7039, Oct. 2013.

[6] P. Ferguson, "Network Ingress Filtering: DDoS Attacks which employ IP Source Address Spoofing," RFC 2827, May 2000.

[7] R. Beverly, A. W. Berger, Y. Hyun, and K. C. Claffy, "Understanding the efficacy of deployed internet source address validation filtering," in *Internet Measurement Conference*, 2009.

[8] J. Li, J. Mirkovic, T. Ehrenkranz, M. Wang, P. L. Reiher, and L. Zhang, "Learning the valid incoming direction of ip packets," *Computer Networks*, vol. 52, pp. 399–417, 2008.

[9] W. Li-Jun, W. Jian-Ping, and X. Ke, "Bgp extension to support inter-domain distributed packets filtering," *Journal of Software*, vol. 18, no. 12, pp. 3048–3059, 2007.

[10] S. Goldberg, "Why is it taking so long to secure internet routing?" *Communications of the ACM*, vol. 57, no. 10, pp. 56–63, 2014.

[11] R. Lychev, S. Goldberg, and M. Schapira, "Bgp security in partial deployment: is the juice worth the squeeze?" in *SIGCOMM*, 2013.
[12] M. Faloutsos, P. Faloutsos, and C. Faloutsos, "On power-law relationships of the internet topology," in *ACM SIGCOMM computer communication review*, vol. 29, no. 4. ACM, 1999, pp. 251–262.
[13] K. Park and H. Lee, "On the effectiveness of route-based packet filtering for distributed dos attack prevention in power-law internets," in *ACM SIGCOMM computer communication review*, vol. 31, no. 4, 2001.
[14] "Peeringdb," http://www.peeringdb.com.
[15] P. Gill, M. Schapira, and S. Goldberg, "Let the market drive deployment: A strategy for transitioning to bgp security," in *ACM SIGCOMM Computer Communication Review*, vol. 41, no. 4, 2011, pp. 14–25.
[16] "As rank by transit degree," http://as-rank.caida.org.
[17] M. Luckie, B. Huffaker, A. Dhamdhere, V. Giotsas *et al.*, "As relationships, customer cones, and validation," in *Proceedings of the conference on Internet measurement conference*. ACM, 2013, pp. 243–256.
[18] "As-relationships," http://data.caida.org/datasets/as-relationships.
[19] "Prefix-to-as mappings," http://data.caida.org/datasets/routing.
[20] B. Quoitin and S. Uhlig, "Modeling the routing of an autonomous system with c-bgp," *IEEE network*, vol. 19, no. 6, pp. 12–19, 2005.
[21] A. Bremler-Barr and H. Levy, "Spoofing prevention method," in *Proceedings IEEE 24th Annual Joint Conference of the IEEE Computer and Communications Societies.*, vol. 1. IEEE, 2005, pp. 536–547.
[22] X. Liu, A. Li, X. Yang, and D. Wetherall, "Passport: Secure and adoptable source authentication," in *NSDI*, 2008.
[23] F. Baker and P. Savola, "Ingress Filtering for Multihomed Networks," RFC 3704, Mar. 2004.
[24] A. Cohen, Y. Gilad, A. Herzberg, and M. Schapira, "Jumpstarting bgp security with path-end validation," in *SIGCOMM*, 2016.
[25] Z. Duan, X. Yuan, and J. Chandrashekar, "Controlling ip spoofing through interdomain packet filters," *IEEE transactions on Dependable and Secure computing*, vol. 5, no. 1, pp. 22–36, 2008.